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Brain Tumor Recurrence vs. Radiation Necrosis Classification and Patient Survivability Prediction

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***Abstract*—GB (Glioblastoma WHO Grade 4) is the most aggressive type of brain tumor in adults that has a short survival rate even after aggressive treatment with surgery and radiation therapy. The changes in magnetic resonance imaging (MRI) for patients with GB after radiotherapy are indicative of either radiation-induced necrosis (RN) or recurrent brain tumor (rBT). Screening for rBT and RN at an early stage is crucial for facilitating faster treatment and better outcomes for the patients. Differentiating rBT from RN is challenging as both may present with similar radiological and clinical characteristics on MRI. Moreover, learning-based rBT versus RN classification using MRI may suffer from class imbalance due to a lack of patient data. While synthetic data generation using generative models has shown promise to address class imbalances, the underlying data representation may be different in synthetic or augmented data. This study proposes computational modeling with statistically rigorous repeated random subsampling to balance the subset sample size for rBT and RN classification. The proposed pipeline includes multiresolution radiomic feature (MRF) extraction followed by feature selection with statistical significance testing (p$<$0.05). The five-fold cross validation results show the proposed model with MRF features classifies rBT from RN with an area under the curve (AUC) of 0.892$\pm$0.055. Moreover, considering the dependence between survival time and censoring time (where patients are not followed up until death), the feasibility of using MRF radiomic features as a non-invasive biomarker to identify patients who are at higher risk of recurrence or radiation necrosis is demonstrated. The cross-validated results show that the MRF model provides the best overall survival prediction with an AUC of 0.77$\pm$0.032. Comparison with state-of-the-art methods suggest the proposed method is effective in RN versus rBT classification and patient survivability prediction.**

***Index Terms*—Classification, radiomics, recurrent brain tumor, radiation necrosis, imbalance data learning, prognostic indexing, survival analysis.**

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I. Introduction

GLIOBLASTOMA (GB WHO Grade 4) is the most common and aggressive primary brain tumor in adults. These tumors are highly invasive and difficult to treat as they spread quickly throughout the brain which poses a significant challenge for medical professionals. Patients with GB (WHO Grade 4) have a poor prognosis even with the standard treatment of maximal safe surgical resection followed by concurrent chemoradiotherapy (CCRT) with temozolomide (TMZ) or adjuvant TMZ. The median overall survival time is only 14–16 months, and the 2-year survival rate is 26–33% [1], [2]. Following chemoradiation treatment, these patients typically undergo regular MR imaging (usually comprising T1WI, T2WI, FLAIR) for monitoring signs of tumor recurrence. A major clinical challenge in evaluating these posttreatment MR images is the differentiation of these lesions as recurrent tumors (rBT) or radiation necrosis (RN). A new enhancement around the resection cavity, progressive enlargement with mass effect suggests tumor recurrence on standard MR imaging. RN is an irreversible radiation-induced injury caused by aggressive radiation treatment and is challenging to diagnose on conventional magnetic resonance imaging (MRI) due to its close visual resemblance to tumor recurrence [3], [4]. The differentiation is further complicated by the simultaneous presence of varying proportions of RN and recurrence/residual tumor confounding the diagnosis on imaging [5], [6].

Currently, the only definitive way to distinguish between RN and rBT is through surgical intervention, followed by a thorough histopathologic evaluation to determine the presence of RN or tumor recurrence in a lesion [7], [8]. Recurrent brain tumors often require aggressive interventions, such as surgery, chemotherapy, or targeted therapies. In contrast, radiation necrosis may necessitate a more conservative approach or even observation [9], [10], [11], [12], [13]. Misdiagnosis can lead to inappropriate treatments, affecting patient outcomes and quality of life. Thus, accurate differentiation is essential for determining appropriate treatment strategies.

Quantitative image descriptors derived from multiparametric MRI are becoming increasingly popular in the biomedical research field as they enhance image-based disease detection and

TABLE I
OVERVIEW OF RELATED WORKS

Previous Work	Aims	Method	Conclusion/Evaluation	Limitations
Prattek et al [17]	To distinguish RN from rBT using radiomics features on routine MRI	Random Forests	Concluded that Random Forests Classifier effectively distinguishes between RN vs rBT and achieved an AUC of 0.79	Only included small, single institution cohorts, and feature selection is parameter dependent.
Park et al. [20]	To establish a high-performing radiomics model with machine learning (ML) from conventional and diffusion MRI to differentiate RN from rBT.	K-Nearest Neighbors, SVM, AdaBoost	Evaluated the ability of conventional and diffusion radiomics to differentiate rBT from RN. The best performance achieved via SVM with ADC images (AUC of 0.80, ACC of 0.78, sensitivity of 0.67, specificity of 0.87)	Underlying representation of data generated by synthetic sampling decrease in model performance
Chen et al. [21]	To identify RN from rBT using multiparametric tissue signature and machine learning	GLM, Random Forests, RDA, SVM	Showed that MRI-based multiparametric radiomics (mpRad) and ML can differentiate RN from rBT in multi-institutional cohort. Random Forest Classifier achieved an ACC of 0.71, sensitivity of 0.52, and specificity of 0.92.	Did not consider underlying correlation between imaging feature and patient's survival data.
* GLM	General linear model	* RDA	Regularized discriminant analysis	

characterization [14], [15]. This method is known as radiomics and involves extracting a number of features from the imaging data [16], [17], [18], [19]. These features are believed to provide insights into the underlying tissue biology. Although several markers have been suggested, their effectiveness as prognostic biomarkers are hindered by the requirement to physically sample the tumor from patients. However, combining potential radiomic features with histopathological findings could enhance clinical decision making without the need for invasive procedures [18], [19].

Many works have attempted to distinguish between rBT and RN by analyzing radiomic features and investigating the challenges posed by imbalanced datasets. These studies have utilized feature selection techniques and classification models like Support Vector Machines (SVM) and Random Forests (RF) [20], [21]. Recently, Prateek et al. [17] have shown the benefits of using a specific set of radiomic features to evaluate efficacy of treatments for brain tumors. These features effectively discriminate between rBT vs RN. However, their approach has limitations due to the need for parameter selection and the presence of high bias in the performance metric. Park et al. [20] use diffusion radiomics from regular MRI and apparent diffusion coefficient (ADC) images to differentiate between recurrent glioblastoma and radiation necrosis. Chen, Xuguang, et al. [21] propose to identify RN from rBT using multiparametric tissue signatures and machine learning. Moreover, these models have several limitations as shown in Table I, including bias due to underrepresentation of minority classes, overfitting, and the inability to generalize the model's performance. Also, these studies only consider a single MRI scan to distinguish RN vs. rBT and do not perform survival analysis to assess the long-term outcomes of patients with RN and rBT. Therefore, further research is needed to investigate the correlation between MR imaging features and survival data through prognostic grouping, aiming to provide more comprehensive insights into these conditions. Additionally, incorporating multimodal MRI scans may enhance the accuracy of distinguishing between RN and rBT by providing a better understanding of the dynamic changes that occur in these tumors.

In addition, identifying patients who are at a low risk of recurrence or radiation necrosis may also potentially lead to more personalized and less aggressive treatment plans, improving the quality of their lives while still ensuring their long-term survival. Radiomics modeling has shown promise in survival analysis [19], [22]. When patient follow-up ends before the time of death, this is called censorship [23], [24], [25]. Censoring time is an important concept in survival analysis as it allows to account for incomplete data and estimate the probability of an event occurring beyond the observed time period. In contrast to simple Cox modeling, which makes a strong independence assumption, copula-based modeling is appropriate if there is a possibility of dependence between survival time and censoring time [26], [27], [28]. Censoring in statistical analysis can potentially introduce bias. By accounting for dependent censoring, copula-based modeling provides more accurate and reliable estimates of survival probabilities and allows for better prognostic grouping of patients.

To address the class imbalance and censoring challenges, this study introduces a statistically robust repeated random subsampling algorithm. The repeated sub-sampling approach encompasses: 1) the utilization of a subject-independent sampling algorithm to generate balanced subsets; 2) the feature selection methodology to identify the most significant features; and 3) the integration of an ensemble tree-based classifier to assess the performance of the model. In the prior research, Leave-One-Out Cross Validation (LOOCV) and stratified 5-fold cross-validation were implemented with an RF classifier to stratify RN and rBT [16]. The study introduces novel computational modeling to overcome class imbalance and dependent censoring challenges to classify between radiation necrosis (RN) and recurrent brain tumors (rBT). Integration of multi-resolution fractal features with traditional radiomic features for multi-institutional data enhances accuracy Furthermore, prognostic indexing is introduced to analyze the correlation between imaging features and patient survivability, marking a significant contribution to computational modeling with multi-institutional data. Leveraging sophisticated multi-resolution fractal features alongside traditional radiomic features, the study aims to distinguish between radiation necrosis (RN) and recurrent brain tumor (rBT) with high precision. Moreover, the research applied copula modeling on extracted imaging features to assess the likelihood of patient survival under dependent censoring scenarios. It focuses on

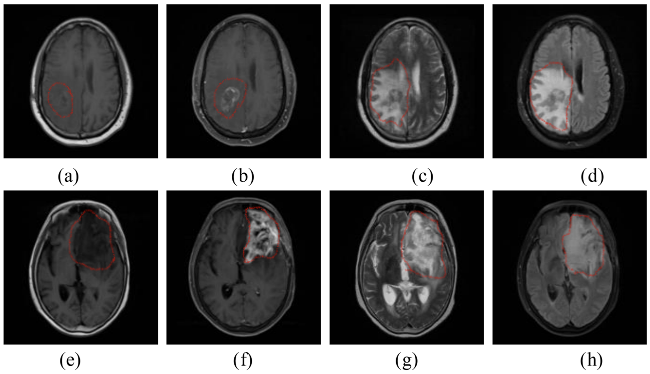

Fig. 1. MRI of two patients with glioma (T1, T1C, T2, FLAIR modalities) (a)–(d) Radiation Necrosis Case after 8 months follow up and Recurrent Case after 6 months follow up (e)–(h).

TABLE II
DEMOGRAPHIC CHARACTERISTICS OF STUDY POPULATION

Data Source	TCGA (rBT=25, RN=0)	25
	TCIA (rBT=104, RN=9)	113
	EVMS (rBT=14, RN=6)	20
Gender	Male	91
	Female	65
	NA	2
Race	African American	5
	White	130
	Asian	1
	NA	22
Age	Min-max	23-80
	Median	58
	Mean	56.35
Survival	Non-censored (rBT=40, RN=4)	44
	Censored (rBT=34, RN=9)	43
	NA	71
Follow ups (months)	rBT	> 6 months
	RN	> 12 months

***NA is unknown, rBT is recurrent brain tumor, RN is radiation necrosis**

predicting patients' survival status, particularly discerning between deceased, and surviving individuals, utilizing statistically significant features (p-value$<$0.05). To evaluate the significance of selected features, a prognostic index (PI) is calculated by aggregating imaging features. The PI effectively categorizes patients into two prognostic groups, enabling differentiation between those at lower and higher risks. The distribution of rBT and RN cases within these prognostic groups is then stratified, revealing the potential of imaging features to categorize rBT from RN within both high and low-risk groups.

II. MATERIALS AND METHODOLOGY

A. Study Population and MRI Images

In this study, the patient data set is collected from the Cancer Genome Atlas (TCGA) in the Genomic Data Commons (GDC) Data Portal [32]; the Cancer Imaging Archive (TCIA) [29], [30], [31], [32], [33]; and in an Institutional Review Board-approved (Eastern Virginia Medical School (EVMS) IRB# 20-6-NH-0130), joint study from collaborating institutions: EVMS, Sentara Healthcare and Old Dominion University (Vision Lab).

A brain tumor recurrence (rBT) occurs when the tumor reappears after treatment, characterized by an increase in enhancing lesion. The radiation-induced necrosis (RN) refers to a focal brain lesion that can occur as a result of various radiation therapy techniques. Radiation necrosis typically manifests at a later stage, typically more than a year after radiation treatment.

However, the presence of contrast enhancement is by no means exclusively indicative of tumor in radiation treated cases and may be indicative of radiation effect and necrosis as well. Fig. 1 presents an example of each of the four different modalities of MRI images showcasing recurrence and radiation necrosis.

In this study, patients with follow-up MRI scans for at least six months for rBT cases and twelve months for RN cases are considered. In all cases, the diagnosis of radiation necrosis and recurrent tumor is confirmed via tissue diagnosis and histopathological confirmation.

A total of 158 patients who underwent surgery followed by radiation treatment are included in this study. Out of the cases, 15 patients are diagnosed with radiation necrosis and 143 patients are identified as having tumor recurrence through histopathological assessment. Following treatment, every patient goes through a series of scans that included T1-weighted pre-contrast (T1), T1-weighted post-contrast (T1C), T2-weighted (T2), and fluid-attenuated inversion recovery (FLAIR). Table II presents the demographic information of patient cases.

B. MRI Preprocessing and Tumor Volume Segmentation

All the patient MRI images have been resampled to 1 mm^3 resolution and undergo skull stripping to remove the skull and non-brain tissues from the MRI images after registering T1, T2, and FLAIR images to T1C images using the linear registration function in FSL5.0.9 (http://fsl.fmrib.ox.ac.uk) [34]. For training the proposed 3D deep learning model, the datasets from RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) Challenge 2021 are used. These datasets are obtained from multiple different institutions under standard clinical conditions [1]. A total of 1251 cases with voxel-level annotations of different tumor sub-tissues: necrotic (NC), peritumoral edema (ED), and enhancing tumor (ET) are used for training the network. The implementation of the deep learning network is done in PyTorch and trains the network on an NVIDIA V100 HPC platform using the BraTS 2021 training dataset (1251 cases) [32], [35]. To train the network, the dataset is randomly split with 80% of data for training and the remaining 20% of the data for validating the trained model. The model trains for 300 iterations. The best version of the segmentation model, tested with validation data, is used for evaluation. The evaluation includes testing with RN and rBT cases. Effective training of the network is observed by the monotonically decreasing loss and the corresponding increase in training dice score as shown in Fig. 2. Tumor volume (ED,ET, NC) is segmented using trained 3D deep learning model [36], [37]. Two experts neuroradiologist (experience more than 10 years) verify the segmented mask to ensure its accuracy. The whole tumor, the tumor core, and the enhanced tumor region are all included in the segmentation used to extract the features.

The subsequent sections describe the feature extraction, feature selection, computational modeling and survival analysis in detail.

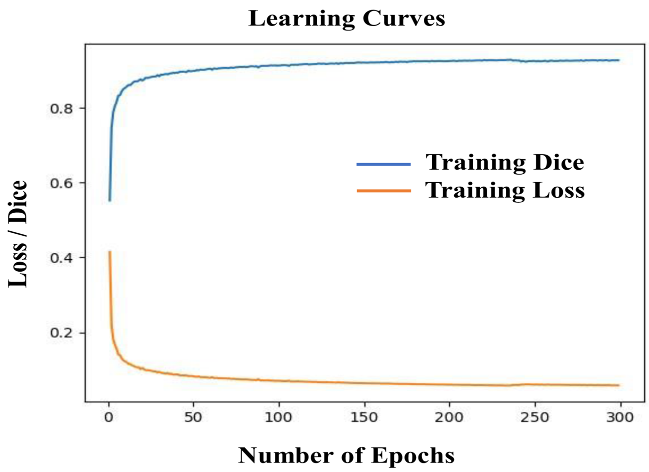

Fig. 2. Learning curves for 3D Brain tumor segmentation model.

C. Feature Extraction

A total of 1620 features have been extracted from the segmented tumor volume. Texture features are extracted from the whole tumor (WT) volumes (including edema, enhance tumor, and necrosis) using MRI techniques such as T1, T1C, T2, and FLAIR. These techniques produce histogram-based statistics and matrix-based features such as the co-occurrence matrix, neighborhood gray tone difference matrix (NGTDM), and the Size Zone Matrix (GLZSM) [18], [19], [38]. The histogram-based statistics from the different tumor sub-regions including enhance tumor (ET), edema (ED), and necrosis (NCR) are also used to extract features from the T1, T1C, T2, and FLAIR MRI sequences (ET, ED, NCR). Volumes of the different regions (ET, ED, NCR) are also described by extracting a number of volumetric features [39], [40], [41], [42]. Finally, the tumor volume is analyzed along three dimensions (x, y, and z) to determine nine spatial features (area, centroid, perimeter, major axis length, minor axis length, eccentricity, orientation, solidity, and extent). Considering the intricate pattern of tumor texture, conventional radiomic feature extraction techniques appear rather homogeneous. The complex texture pattern of brain tumor tissues in MRI can be captured by fractal dimension (FD) that is mathematically estimated by multifractional Brownian motion (mBm), piecewise triangular prism surface area (PTPSA) and multi-resolution fractal wavelet model [43], [44]. These three algorithms are utilized to extract texture variation by characterizing fractal dimension in the tumor region of MRI. The mBm model effectively captures spatially varying heterogeneous tumor texture. It also enables the capture of spatially varying random inhomogeneous tumor texture at different scales.

D. Feature Selection and Computational Modeling

The algorithm below illustrates the framework for feature selection for the features extracted from the tumor volumes. The process of feature selection involves two steps: (1) training a decision tree classifier and calculating the F1 score for each feature over 1000 iterations with 5-fold cross validation; and (2) selecting the features that are most statistically significant (with a p-value$<$0.05) after applying a threshold that is based on F1 score. The user defined threshold (0.85 for MRF and 0.80 for NFRF) selects the statistically significant features. A total of 8 out of 30 and 7 out of 28 significant features have been selected for MRF and NFRF model, respectively. The computational framework for repeated random sampling with feature selection is shown in Fig. 3.

Algorithm 1: Repeated Random Sampling for Feature Ranking.

Step 1: Initialization: number of iterations i, number of folds n, feature data frame D_T, majority samples N_1, minority samples N_0
Step 2: Define $N_1^{'}$ as 15 patients randomly sampled from 143 from majority class N_1 in each iteration i, N_0 15 patients with minority class in each iteration i, y as target class vector and $\{ X_j\}, j = 12, \ldots\ldots, k$ the feature matrix in the data frame D_T, k is number of total features.
Step 3: for iteration $i = 0$ to i-1
  Initialize $N_1^{'}$, $N_2^{'} = N_0$, $D_T^{'} = N_1^{'} \cup N_2^{'}$,
  Store Subsample: $D_T^{'}$ after each i_{th}iteration
  within $D_T^{'}$ split the target variable vector y and feature variable matrix as $\{X_j\}, j = 12, \ldots\ldots, k$
Step 4: for fold $n = 0$ to n-1 do
    enumerate train and test indices for n_{th} fold in $\{X_j\}$, $j = 12, \ldots\ldots, k$ and y
Step 5: for $j = 0$ to k-1 do
  fit a decision tree classifier on the train indices of $\{X_j\}$
  predict $\hat{y}$ on the test indices of $\{X_j\}$
  calculate F1-score based on true y and predicted $\hat{y}$ of the test indices of $\{X_j\}$
  assign the score as the feature score of each $\{X_j\}$
  end for
    end for
    Return: Cumulative F1-score after n-fold cross-validation
  end for
Step 6: Output: Features ranking based on cumulative score after i iteration.

A computational model with repeated random sampling is applied to automatically predict rBT and RN cases. The proposed method employs 1000 iterations, random sampling within each iteration followed by 5-fold cross-validation. Random sampling is a potential solution to handle imbalance in rBT and RN data set Algorithms 1 and 2. The discriminative features are used to assess the effectiveness of both multiresolution fractal features and conventional radiomics features. This study compared two different model configurations. The model configurations are a non-fractal model (NFRF), which includes only conventional volume, area, and texture features, and a fractal model (MRF), which incorporates multi-resolution fractal features along with conventional volume, area, and texture features.

Using this repeated sampling method, the training data is partitioned into sub-samples, with each sub-sample containing an equal number of instances from each class. This sampling

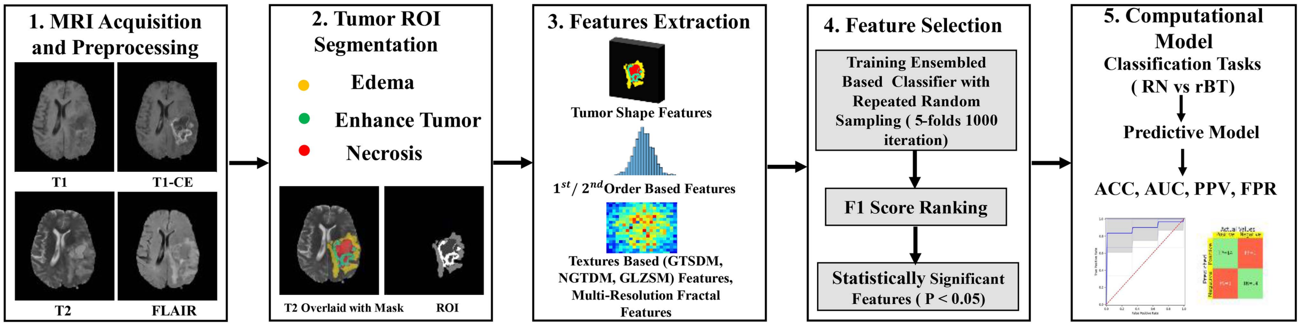

Fig. 3. Overall pipeline of the proposed method for rBT vs RN classification with Repeated Random Sampling Computational Model.

Algorithm 2: Repeated Random Sampling n-Fold Cross Validation With i_{th} Iteration.

Step 1: Initialization: $D_T{}'$ after each iteration i, number of folds n, $\{X_j\}$, $j = 12, \ldots\ldots, k$ feature matrix
Define $\{X_{select}\} \subseteq \{X_j\}, j = 12, \ldots\ldots, k$, based on ranking score, classification model C
Step 3: for iteration $= 0$ to -1 do
Step 2: Initialization: $D_T{}' = N_1{}' \cup N_2{}'$, $\{X_{select}\}$ and target vector y from $D_T{}'$
Step 4: for fold $n = 0$ to n-1 do
  enumerate train and test indices for n_{th}-fold in $\{X_{select}\}$ and $\{y_{select}\}$
  fit a classifier on the train indices of $\{X_{select}\}$ and $\{y_{select}\}$
    validate classifier on the test index of $\{X_{select}\}$ and $\{y_{select}\}$
end for
  Return: Cumulative performance metrics across all n for this iteration
end for
Step 5: Return: Average the performance metrics across all iterations to obtain the final cross-validated.

approach is used to ensure that each sub-sample is representative of the overall population. Only the most relevant features are used in this modeling process. The purpose of cross-validation is to ensure a dependable estimation of a model's performance on data that has not been seen before. It is useful for evaluating the model's efficiency and ability to apply to different situations. Through the process of dividing the data into k subgroups with similar sizes, k-fold cross-validation guarantees the model's resilience. In addition, it is standard procedure to determine the sample size (n) by multiplying k, the number of folds, with m, the number of observations in each fold or subset. Given a dataset with 158 patient cases, divided into 5 folds with 30 observations each, the total number of cases is calculated by multiplying 5 and 30. Given the circumstances, it appears to be a logical decision to employ a 5-fold cross-validation method for dataset partitioning.

Algorithm 2 shows the training procedure of a classifier for $\boldsymbol{n}$-fold cross validation with $\boldsymbol{i}$ iterations**.** Using gradient boosting decision trees, CatBoost [45], [46] which is an ensemble method for machine learning. CatBoost employs gradient boosting, a method that sequentially improves prediction accuracy by minimizing errors made by earlier decision trees within the ensemble. In contrast, Random Forest (RF) comprises decision trees that are individually trained on bootstrapped sub-samples of the dataset, without consideration for the performance of other trees in the ensemble. This difference in methodology allows CatBoost to adaptively adjust its predictions based on the collective knowledge of the ensemble.

E. Survival Analysis Modeling

Survival analysis often uses time-to-death event times. The best quantitative indicator of cancer patients' well-being is overall survival (OS). Survival analysis is made more difficult by censorship. When a study stops taking new patients or when a patient drops out, it omits their survival data. When examining survival data, medical researchers and statisticians typically make the assumption that censoring mechanisms are unrelated to the event of interest. Statistical independence between events and censoring times is taken for granted by survival analysis methods like the Kaplan-Meier estimator and Cox regression.

Considering two random variables where, survival time T refers to the length of time of an individual or group survives from a specific event or diagnosis and censoring time U is the point at which data collection stops, often due to the end of a study or loss to follow-up. It is assumed that survival time T and censoring time U are conditionally independent given a feature x_j for all j = 1 …., p. Rather than assuming complete independence, it is more realistic to assume independence only under certain conditions.

Let's consider T is the amount of time spent alive, U is the amount of time spent being censored, and x is a vector of censoring time T and U are assumed to be conditionally independent if and only if there is no correlation between them and any of the components of x. This assumption is typically made for Cox regression models. Also, let the marginal survival functions given x be denoted by $S_T\ (t|x) = P_r\ (T > t|x)$ and $S_U\ (u|x) = P_r\ (U > u|x)$ where the probability is given as,

$$P_r\,(T > t, U >\ u|\,\mathbb{X}) = C_\theta\ \{S_T\,(t|\,\mathbb{X})\,,\ S_U\,(u|\,\mathbb{X})\}\,; \quad (1)$$

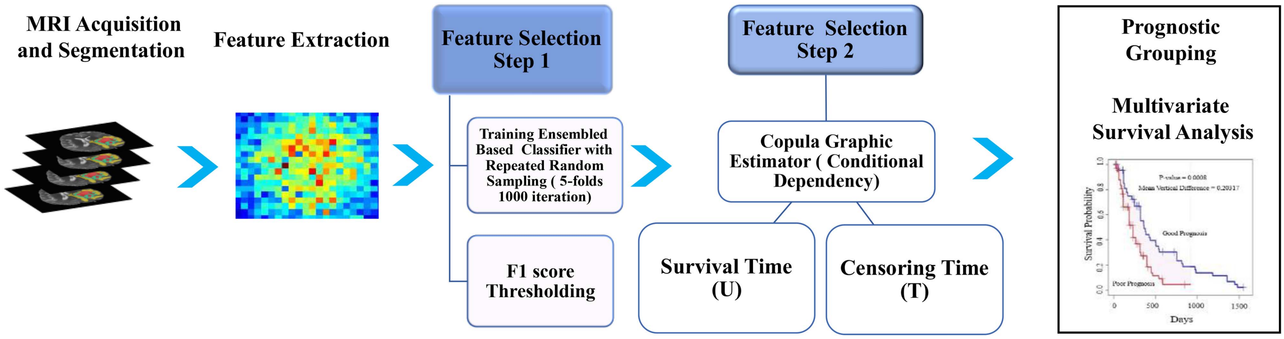

Fig. 4. Overall pipeline for Survival analysis.

TABLE III
DISTRIBUTION OF PATIENTS FOR SURVIVABILITY ANALYSIS

Patient Status	Passed Away	Alive
Non-Censored Patient =44	44 (rBT=40, RN=4)	0
Lost follow up /Censored =43	32 (rBT=32, RN=0)	11 (rBT=2, RN=9)

where C_θ is a copula function [23], [26] and θ is a parameter characterizing the degree of dependence between T and U. Here, a bivariate survival function is taken into consideration. According to this model, the dependency between T and U is characterized by the variable C_θ .The copula function illustrates the interdependence of the variables represented by the marginal distributions of survival time and censoring time. The correlation between two variables, T and U, can be evaluated using Kendall's τ (s), which is defined as,

$$\tau = P_r\left\{(T_2 - T_1)(U_2 - U_1) > 0 | \mathbb{X}\right\} - P_r\left\{(T_2 - T_1)(U_2 - U_1) < 0 | \mathbb{X}\right\}; \quad (2)$$

where (T_1, U_1) and (T_2, U_2) are values derived from the model (1), respectively.

A copula that captures the dependence pattern in the data is suitable for survival analysis. It is known that different copulas imply different types of dependence of survival functions. For example, the Clayton copula is known to impose dependence for lower survival and censoring probabilities (i.e., longer survival and censoring times). In contrast, the Joe copula imposes the dependence of survival function for larger survival and censoring probabilities (i.e., shorter survival and censoring times) [24], [27]. Considering the dependent censoring scenario presented in Table III which happens mostly for lower survival times, the clayton copula is used for feature selection.

Fig. 4 illustrates the complete pipeline for survival analysis. It incorporates two steps for feature selection by taking into account conditional dependency. Initially, the dependency parameter is determined by analyzing the survival data and the radiomics features matrix. The statistically significant features (p-value < 0.05) are then used to make binary predictions about the patient's survival status.

III. RESULTS AND DISCUSSION

This section discusses two representative experiments in this study as follows.

A. Experiment 1: Classification of Recurrent Brain Tumor From Radiation Necrosis in GBs

In this experiment, the relative importance of the features is calculated first using corresponding F1 score, and the results are ranked. To find the statistically significant features, independent t-tests are performed for normally distributed continuous variables, and Mann–Whitney U-tests for continuous variables without normal distribution. Statistical analysis shows that 8 (P < 0.05) out of the 27 and 7 (P < 0.05) out of 30 are the most important features from each of the MRF and NFRF methods, respectively.

Initially, the efficacy of conventional radiomics and multiresolution fractal features extracted from multimodality MRI images has been evaluated. Table IV shows that the classification performance metrics of multimodality MRI is higher in MRF model than that of NFRF model, which reveals the usefulness of employing fractal features extracted from different MRI modalities. In order to better reflect the class imbalance due to underrepresentation of minority cases and improve the performance of radiomics, some scholars have proposed the use of synthetic sample generation. Park et al. [20] evaluated the ability of conventional feature and diffusion radiomics to differentiate rBT from RN. The performance was increased when synthetic minority over-sampling technique (SMOTE) was applied in minority dataset. Xuguang, et al. [21] evaluated random oversampling and SMOTE during model optimization to circumvent the potential model instability caused by class imbalance. Finally, they found that a random forest model trained with radiomics tissue signature and random oversampling achieved the highest AUC in the validation cohort.

In this study, the repeated random resampling method is used due to the small size of the dataset (Algorithm 1). The AUC, accuracy, positive predictive value (PPV), and false positive rate (FPR) values obtained over 1000 iterations of a 5 fold-cross validation via a decision tree classifier. Fig. 5 shows the ROC curves and confusion matrix (single iteration) of the classification of RN versus rBT for MRF and NFRF model. This shows that the

TABLE IV
COMPARISON OF CLASSIFICATION PERFORMANCE OF THE PROPOSED FRAMEWORK WITH OTHER METHODS FOR RECURRENT TUMOR VS RADIATION NECROSIS CLASSIFICATION

Model Configurations	Method	Dataset size	Area Under Curve (AUC)	Accuracy (%)	Positive Predicted Value (PPV)	False Positive Rate (FPR)
Proposed MRF Model	**Repeated Random Sampling**	**RN=143, rBT=15**	**0.89±0.055**	**80.88±0.061**	**0.80±0.064**	**0.15±0.07**
Proposed NFRF Model	**Repeated Random Sampling**	**RN=143, rBT=15**	**0.86±0.061**	**77.43±0.064**	**0.77±0.064**	**0.16±0.08**
IBLR	**0.632 Bootstrap**	**RN=143, rBT=15**	**0.84±0.012**	**-**	**-**	**-**
SMOTE-MRF	**SMOTE**	**RN=143, rBT=15**	**0.835**	**91.00**	**0.73**	**-**
ADASYN-MRF	**ADASYN**	**RN=143, rBT=15**	**0.85**	**80.00**	**0.63**	**-**
Random Forest (Xuguang, et al.[21])	SMOTE	RN=37, rBT(TP)=98	0.71	-	-	-
Random Forest (Prateek, et al. [20])	-	RN=22, rBT=33	0.79	-	-	-
SMOTE (Park, *et al*. [17])	SMOTE	RN=18, rBT=23	0.80	78.00	-	-

The bold values are our experimental results which we have compared with others in the literature.

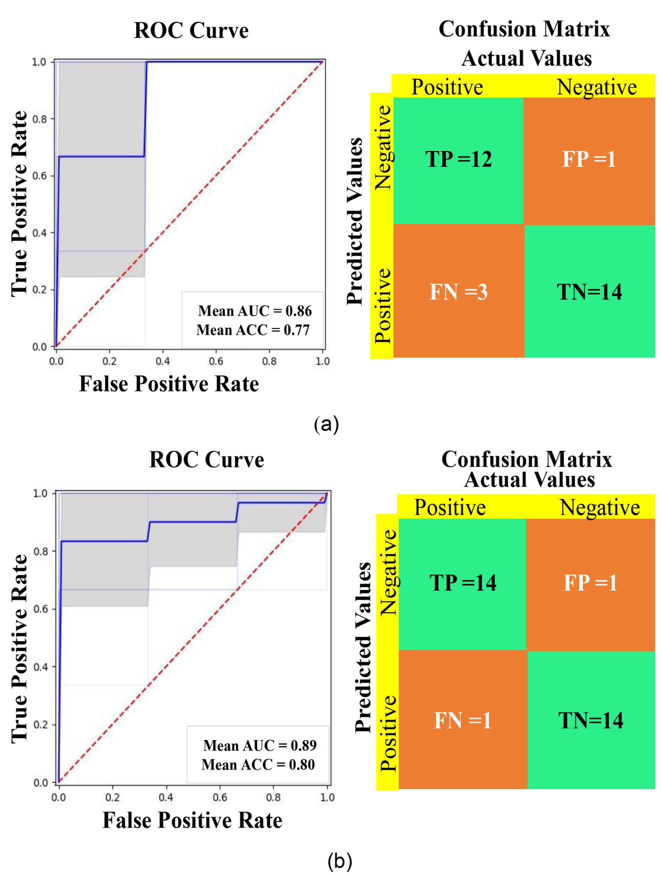

Fig. 5. ROC curve and confusion matrix with mean (AUC, ACC) for (a) MRF model and (b) NFRF model.

variation tendency of the ROC of the is the same across the folds, and the difference between the AUC values is very small, which indicates that overfitting on the proposed method is alleviated in this study.

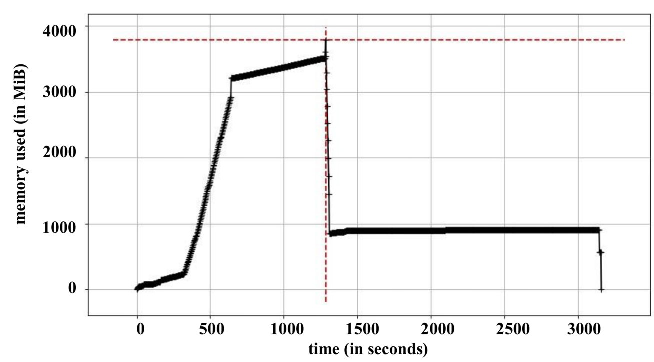

Fig. 6. Memory usage plot over time for the proposed models.

For comparison with oversampling methods SMOTE and ADASYN are implemented for the classification task. Table IV reveals that the classification accuracy of using SMOTE and ADASYN [20] is lower than that of employing repeated random sampling in the classification of RN versus rBT. Moreover, bootstrap resampling with imbalance adjustment learning (IBLR) method is also employed to compare the performance of the proposed model. Regardless of the performance of oversampling, SMOTE and ADASYN are less stable in classifying rBT vs RN. Finally, the proposed MRF model obtaining an AUC of 0.89±0.055, the Positive Predicted Value (PPV) of 0.80±0.064, and the False Positive Rate (FPR) of 0.15±0.074 indicates the effectiveness in classifying rBT vs RN.

The memory profiler package [47] is used to visualize the memory usage, which allows for profiling memory usage by monitoring memory consumption at each line of code. This enables the identification of memory leaks and code optimization to enhance performance. The Fig. 6 illustrates the memory usage changes over time during the implementation of the model.

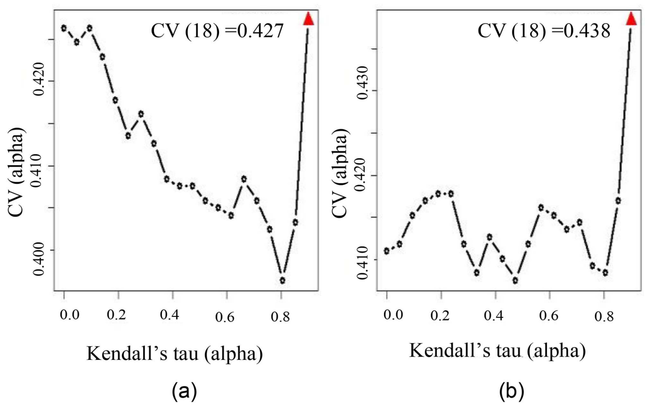

Fig. 7. The cross validated c-index for the MRF and NFRF models. The maximum value of the cross-validated c-index can be found at $\alpha = 18$ which corresponds to Kendall's tau = 0.90. (a) c-index = 0.427 for MRF model, (b) c-index = 0.438 for NFRF model.

B. Experiment 2: Survival Prediction and Patient Outcome

In this study, out of the total of 158 patients, 44 have passed away, 43 have been lost to follow-up (32 passed away and 11 alive), and 114 have neither. Therefore, a total of 87 (44 + 43) patients are used for the survival analysis and prediction. In case of censorship or lack of follow-up meant that those patients are excluded from survival analysis.

The standard practice for selecting important features for survival analysis without considering the dependency between survival and censoring time [21] is to use a univariate simple Cox proportional hazard model. For survival analysis the copula method is employed for feature selection in the censoring scenario. The feature matrix for these 87 patients is written as $X_i = (x_{i1}, x_{i2}, \ldots, x_i p)$. Then, the significant features (p-value < 0.05) are used to predict the probability of survival outcomes and to make a binary prediction of whether the patient is expired or dead (denoted as 1) or not expired or dead (denoted as 0).

The strategy proposed here aims to consider the impact of dependent censoring on biomarkers that may potentially affect the marginal survival. The experiments and statistical significance analysis demonstrate that the selected features may be used to develop a predictive model for patient outcomes. A plot of the cross-validated c-index utilizing the training samples is presented in Fig. 7. The c-index is at its highest point when the association parameter is equal at $\theta = 18$ which corresponds to Kendall's tau = 0.90. In dependent censoring, the c-index is physically significant because it quantifies a model's discrimination ability relative to the risk of the event of interest. Better discrimination between subjects with different risks of experiencing the event of interest is indicated by a higher c-index.

The marginal survival curves for the good (or poor) prognosis are shown in Fig. 8 for MRF and NFRF model. To correct for the inherent bias in the Kaplan-Meier survival curve under independent censoring scenario, the copula-graphic estimator of the survival curves is used which has been modified by the Clayton Copula at $\theta = 18$ (Kendall's tau = 90). The average vertical difference between the survival curves over the study period is used to determine the degree of separation between the curves, and the corresponding p-value is calculated using the permutation test. Smaller p-values correspond to a better separation of the patients with a good prognosis from those with a poor prognosis. In dependent censoring scenario, the proposed method produces significant separation between the patients with a good prognosis and those with a poor prognosis (p-value < 0.05) while using multivariate analyses (dependent censoring).

According to the findings, the MRF model is effective at determining which features are most important for making accurate prognostic stratification about patient outcomes. The coefficient value obtained from radiomic features in the dependent censoring setting is statistically significant. In this case, the copula feature selection (multivariate) method provides an extremely clear separation between the good and poor prognosis patients (p-value = 0.008 as shown in Fig. 8(a)). In case of NFRF model, the copula feature selection (multivariate) method leads to a better separation of the good and poor prognoses (p-value = 0.001; Fig. 8(c)) compared to that for independent scenario (p-value = 0.2199; Fig. 8(d)). Fig. 8 shows that, in contrast to the case of independent scenario ($\theta = 0$, it becomes simple cox model), the MRF and NFRF perform significantly better at distinguishing between good and poor prognoses when subjected to dependent censoring. Hence, the list of features selected by the multivariate copula method is consistently more predictive than those selected under independent scenario which is simple Cox model. This suggests that the proposed methods are more effective in identifying potential radiomic biomarkers that are associated with good and poor prognosis group. Therefore, they may be useful in developing more accurate treatment plans to significantly improve the diagnosis and management of recurrent brain tumors and radiation necrosis.

To assess the significance of the selected features, a prognostic index (PI) is calculated by linearly combining these features. This PI is then used to categorize patients into prognostic groups, differentiating between those at lower and higher risks. Furthermore, the distribution of cases within these prognostic groups is examined, distinguishing between RN and rBT cases. The PI values are commonly used in clinical practice to aid decision-making and to stratify patients into risk groups. They can also be useful in designing treatment planning by identifying patients who are more likely to benefit from a treatment. The number of patients with a good or poor prognosis is identified using the prognostic indices as shown in Fig. 9. PI is high for poor prognosis/high risk groups and PI is low for good prognosis/low risk groups. This information can help us to stratify patients' groups who may require more intensive treatment or monitoring and those who may have a better chance of recovery. It also highlights the importance of early detection and intervention in improving patient outcomes.

Additionally, 61.37% of rBT cases fall into the bad prognostic category in the fractal model, as depicted in Fig. 9(a). In the non-fractal model, 59.3% of rBT cases belong into the bad prognostic category, as shown in Fig. 9(b) Within the bad prognostic group, the proportion of rBT cases is greater for the fractal model

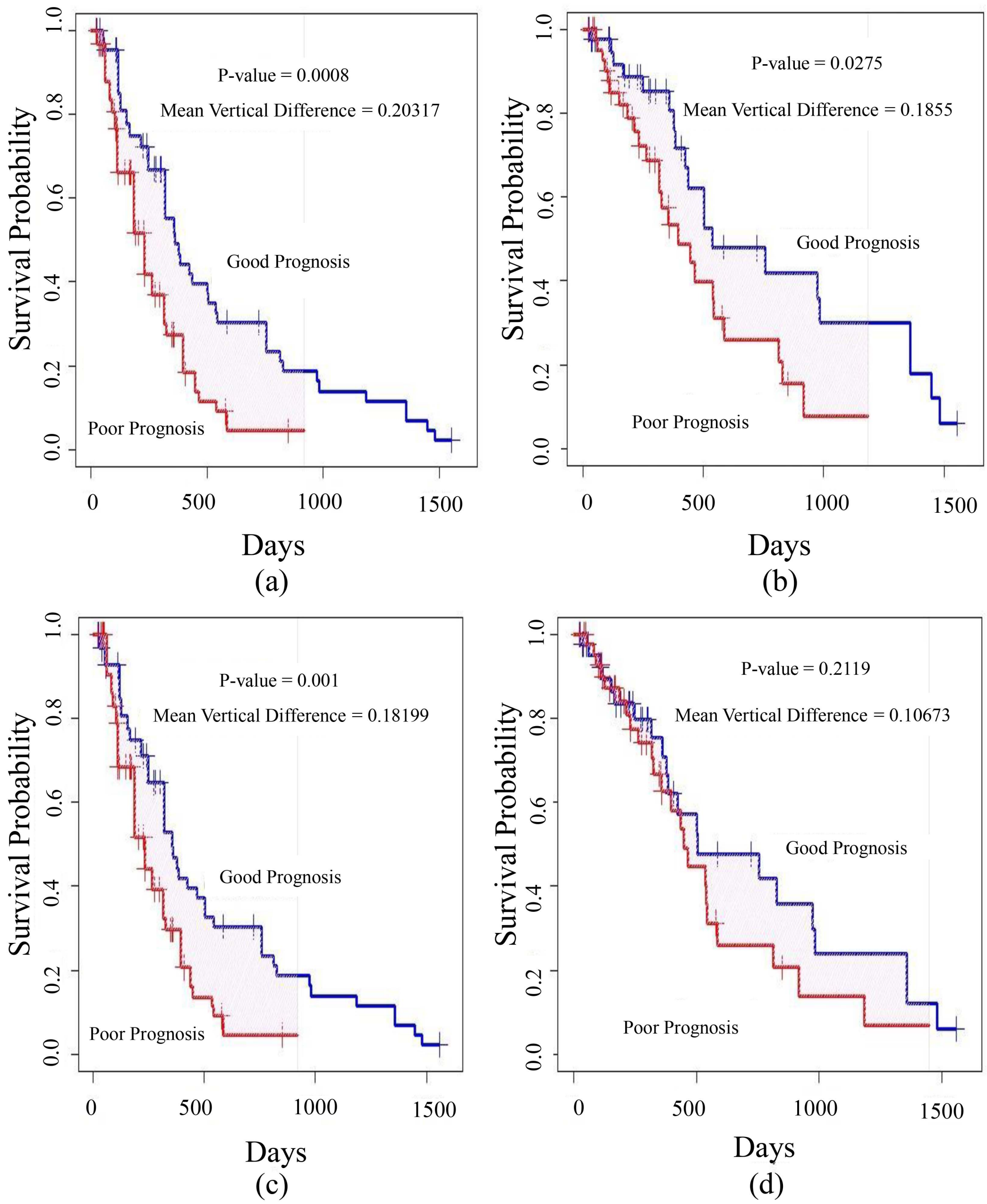

Fig. 8. The marginal survival curves for the group with a good (or poor) prognosis determined by the most significant features. MRF model: (a) Dependent censoring (multivariate), (b) Independent censoring (simple Cox model). NFRF model: (c) Dependent censoring (multivariate), (d) Independent censoring (simple Cox model).

compared to the non-fractal model. It is evident from the plot that most patients in rBT have a high prognostic index value which are associated with worse survival outcomes. This means that they are more likely to experience disease recurrence, radiation necrosis, or death compared to patients with a low score. In this case, a high prognostic index value could be used to identify patients who may require for targeted interventions to improve their outcomes.

From the analysis, it is observed how dependent censoring impacts feature selection and survival probability. To predict survival status, the features that are statistically significant (with a p-value < 0.05) for both the fractal and non-fractal models are used in different combination (35,7 and 10).

Table V displays the predictive outcomes of a 5-fold cross-validation with 1000 iterations for binary survival prediction.. Regarding binary survival prediction, the accuracy of the models is emphasized for the selected features. Table V shows that a lower false positive rate is correlated with higher precision. It is observed that, using the top 7 significant features both MRF and NFRF models have high positive predictive value. When comparing the performance of the models, the MRF model is found to be more statistically significant than the NFRF model.

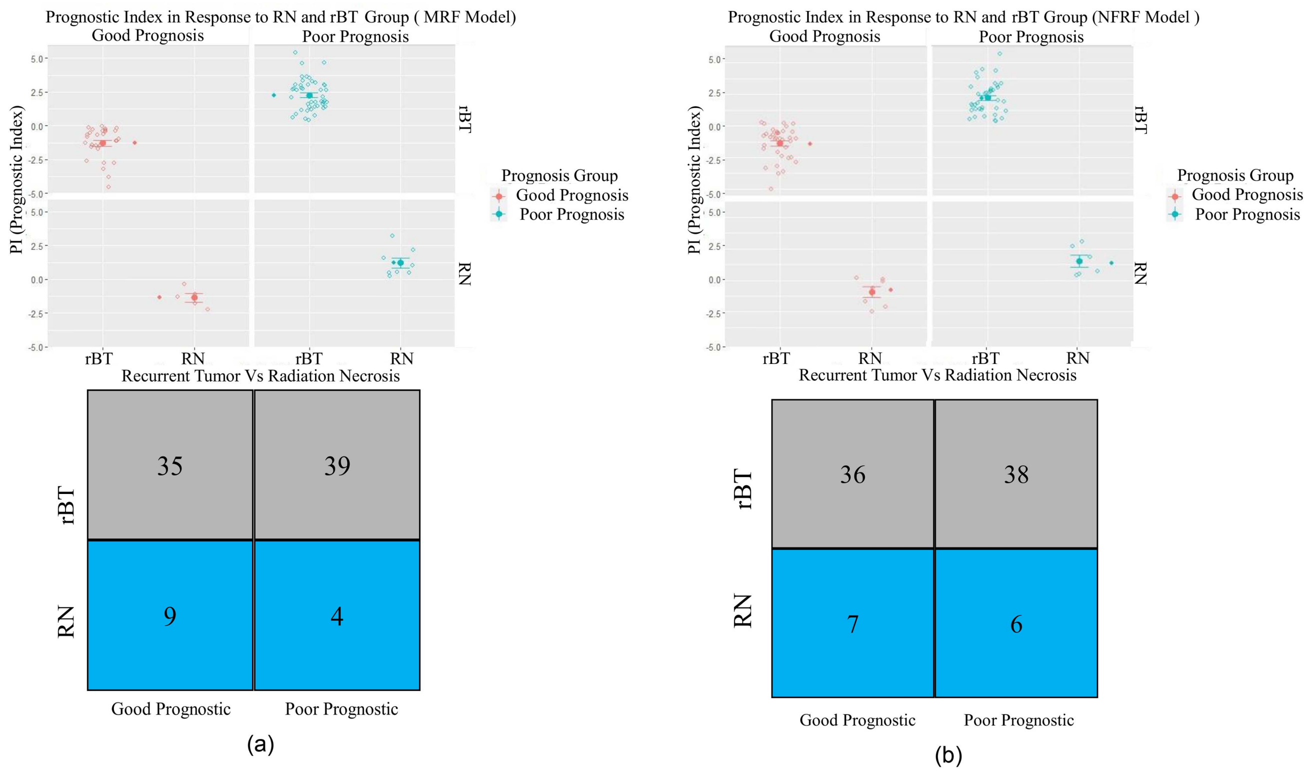

Fig. 9. Two-way plot for prognostic index in response to good or poor prognosis: (a) MRF model, (b) NFRF Model.

TABLE V
COMPARISON OF DIFFERENT CONFIGURATIONS OF THE MODEL BASED ON THE MEAN TEST PERFORMANCE ACROSS $n = 5$-FOLD WITH $i = 1000$ ITERATIONS

Number Of features	Models	Area Under Curve (AUC)	Accuracy (%)	Positive Predicted Value (PPV)	False Positive Rate (FPR)
Top 3 features	MRF	0.73±0.035	72.61±0.027	0.62±0.033	0.34±0.046
	NFRF	0.67±0.036	70.77±0.026	0.60±0.032	0.36±0.043
Top 5 features	MRF	**0.77±0.032**	**76.98±0.025**	**0.68±0.030**	**0.35±0.041**
	NFRF	0.73±0.035	71.99±0.026	0.61±0.034	0.35±0.048
Top 7 features	MRF	0.77±0.032	76.91±0.025	068±0.031	0.27±0.041
	NFRF	0.76±0.032	75.99±0.025	0.66±0.033	0.29±0.042
Top 10 features	MRF	0.75±0.033	76.21±0.025	0.66±0.033	0.28±0.042
	NFRF	**0.75±0.033**	**76.54±0.026**	**0.67±0.034**	**0.28±0.044**

The bold values are the highest accuracy of our proposed method.

IV. CONCLUSION

In conclusion, discovering a noninvasive computational modeling approach to differentiate between rBT and RN holds great clinical importance. This study investigates a new computational framework that combines multiresolution features from multi-modality MRI images to handle imbalance data from multiple institutions. The sophisticated fractal features alongside traditional radiomics features have shown the potential to greatly benefit precision medicine and may enhance treatment strategies for rBT and RN. There are some limitations to the proposed method. In addition, the study's sample size is rather small, which could restrict the wide applicability of these findings. Efforts have been made to address these limitations. Random sampling with 5-fold cross-validation is utilized for feature selection and model assessment to ensure a balanced representation of patient cases in every iteration. Furthermore, only significant features have been included in classification and survival modeling. Survival analysis utilizes copula modeling to address the issue of independent censoring assumptions. Table IV shows that the classification results of the proposed method outperform the results of recently published papers. Here, only the results of different methods in the literature are listed. Direct comparison of the performances of different methods is unreasonable because various datasets and methods for extracting features and building classifiers are used among studies. Nonetheless, the proposed method shows the highest AUC, specificity, and accuracy among the methods surveyed for the classification problem, thereby implying its advantage for classifying rBT and RN.

Additionally, potential biomarker selection strategies for survival data with dependent censoring have been evaluated. Identifying the most potential radiomic features, Copula model multivariate (dependent censoring) feature selection methods found suitable to handle the bias of conditional dependency. The results and simulations of this study show that the proposed methods achieved improve performance for finding potential radiomic biomarkers associated with good and poor prognoses when dependent censoring is present. For future work, the proposed model needs to be validated using larger patient cohorts.

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